

CS201 Midterm 1 Part A

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I. QUESTION (ALGORITHM ANALYSIS) (20 POINTS)

```

1 int algorithm1(int N){
2     int i , j , k, sum = 0;
3     for ( i = 1; i <= N; i++)
4         for ( j = i ; j <= N; j++)
5             for (k = 1; k <= j - 2 * i; k++)
6                 sum++;
7
8     return sum;
9 }
```

What does the function algorithm1 return in terms of N? You may assume N is power of 2.

II. QUESTION (ALGORITHM ANALYSIS) (20 POINTS)

```

1 int algorithm2(int N){
2     if (N == 0)
3         return 0;
4     int sum = 0;
5     for (int i = 0; i < N; i++)
6         for (int j = 0; j < N; j++)
7             sum++;
8     for (int i = N; i > 0; i--)
9         for (int j = 1; j < N; j *= 2)
10            for (int k = 1; k < N; k *= 2)
11                sum++;
12     for (int i = 0; i < 4; i++)
13         for (int j = 0; j < 6; j++)
14             if ((i * j) % 3 == 1)
15                 sum += algorithm2(N / 2);
16     return sum;
17 }
```

What is the time complexity of algorithm2? You must construct the recurrence equation and solve it without using the master theorem.